

Hypertensive Encephalopathy

Section 1: Case Summary

Scenario Title:	Hypertensive Encephalopathy
Keywords:	Delirium, hypertensive emergency
Brief Description of Case:	This case involves the assessment and management of an acutely confused patient with hypertensive encephalopathy. The resident is expected to initiate appropriate supportive measures (ensure working IV, oxygen, monitors) while obtaining a focused history and performing a focused physical exam, including a detailed neurologic assessment. They should recognize the elevated BP and order a work-up for end-organ damage. The patient will require transfer to a critical care setting for administration of acute BP lowering agents including involvement of RACE or a similar high acuity response team.

Goals and Objectives	
Educational Goal:	To practice the initial assessment and management of a hypertensive emergency
Objectives: (Medical and CRM)	1. Review the initial approach to the unstable floor patient 2. Work through the differential for acute confusion 3. Recognize and initiate initial management of hypertensive emergency 4. Demonstrate an organized management plan (IV, O2, monitors, call for help, consider BP lowering therapies) 5. Initiate work-up for the underlying etiology of the hypertensive emergency
EPAs Assessed:	

Learners, Setting and Personnel			
Target Learners:	☒ Junior Learners	☐ Senior Learners	☐ Staff
	☒ Physicians	☐ Nurses	☐ RTs
	☐ Other Learners:		
Location:	☒ Sim Lab	☐ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1		
	Sim Actors: 1		
	Sim Techs: 1		

Scenario Development	
Date of Development:	2015
Scenario Developer(s):	Dr. Tim Chaplin
Affiliations/Institutions(s):	Queen's University
Contact E-mail:	chaplintim2@gmail.com
Last Revision Date:	Jan 2020
Revised By:	Dr. Chris Heyd
Version Number:	1



Hypertensive Encephalopathy

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Micah Brown		Age: 73		Gender: M	Weight: 90kg
Presenting complaint: Confusion					
Temp: 37.2 °C	HR: 88	BP: 227/134	RR: 22	O ₂ Sat: 94%	FiO ₂ : Room air
Cap glucose: 6.6			GCS: 14 (E4 V4 M6)		
Triage note: It is 3:00 am, you are called to the floor to assess a 73M, Mr. Brown, who was admitted to general surgery 2 days ago, and has a scheduled large bowel resection for diverticulitis tomorrow. He has been undergoing bowel prep. You are called because he is experiencing confusion, shortness of breath, and chest heaviness.					
Allergies: None					
Past Medical History: Prostatectomy 5 years ago for prostate cancer Frequent diverticulitis flares requiring antibiotics			Current Medications: coversyl 4mg BID metoprolol 50mg BID hydrochlorothiazide 25mg daily multivitamin		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	
RN to provide when asked: This is your third shift with MR. Brown. His baseline is very pleasant, alert and oriented. You found him this morning confused, appearing unwell (flushed, diaphoretic) and you have called the resident to assess Mr. Brown.	
At 2-3 minutes, nurse will report that regular anti-hypertensives have been held as a pre-op strategy.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Normal	Neuro: Confused, slow to answer, rotary nystagmus
Resp: Normal	Head & Neck: Normal
Abdo: Diffuse tenderness	MSK/skin: Diaphoretic
Other: Appears unwell	

Hypertensive Encephalopathy

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin Adult
☒ Standardized Patient (If possible, will add to the realism)
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Vitals machine Manual BP cuff
C. Required Medications
None
D. Moulage
Diaphoresis Hospital Gown IV in situ
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i>
Pale, diaphoretic, unwell Disoriented, slow to answer Rotary nystagmus Moans to palpation of the abdomen

Hypertensive Encephalopathy

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
	RN NOTES • This is your third shift with MR. Brown. His baseline is very pleasant, alert and oriented. You found him this morning confused, appearing unwell (flushed, diaphoretic) and you have called the resident to assess Mr. Brown. • If asked, you are not able to give IV BP meds on the floor (RACE may do this for a brief time). • At 2-3 minutes nurse will report that regular anti-hypertensives have been held as a pre-op strategy.

Hypertensive Encephalopathy

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus HR: 88 BP: 227/134 RR: 22 O ₂ SAT: 94% T: 37.2°C GCS: 14 (E4 V4 M6)	Disoriented, slow to answer	<u>Expected Learner Actions</u> ☐ History and Physical ☐ Neuro exam ☐ Full set of vital signs ☐ Cap sugar ☐ EKG ☐ Review medications ☐ Send bloodwork ☐ Supplemental O ₂ ☐ Call for senior surg resident	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - - <u>Triggers</u> <i>For progression to next state</i> All actions complete or 3-5 mins elapsed → 2. Unchanged	
2. Unchanged Rhythm: Sinus HR: 88 BP: 234/138 RR: 20 O ₂ SAT: 99% (2L NP) GCS: 14 (E4 V4 M6)	Same	<u>Expected Learner Actions</u> ☐ Arrange CT head ☐ Consider anti-HTN agent ☐ Serial BP checks ☐ Call RACE/medicine	<u>Modifiers</u> Give oral anti-HTN → no change If requests IV antihypertensives → RN to prompt “we cannot give IV without high acuity team (RACE)” <u>Triggers</u> All actions complete → 3. Results Back	
3. Results Back HR: 88 BP: 234/138 RR: 20 O ₂ SAT: 99% (2L NP) GCS: 14 (E4 V4 M6)	Same	<u>Expected Learner Actions</u> ☐ Review bloodwork ☐ Review CT scan ☐ Arrange higher level of care	<u>Modifiers</u> <u>Triggers</u> Discuss transfer with RACE → End Case	Give Bloodwork and CT Results at beginning of state



Hypertensive Encephalopathy

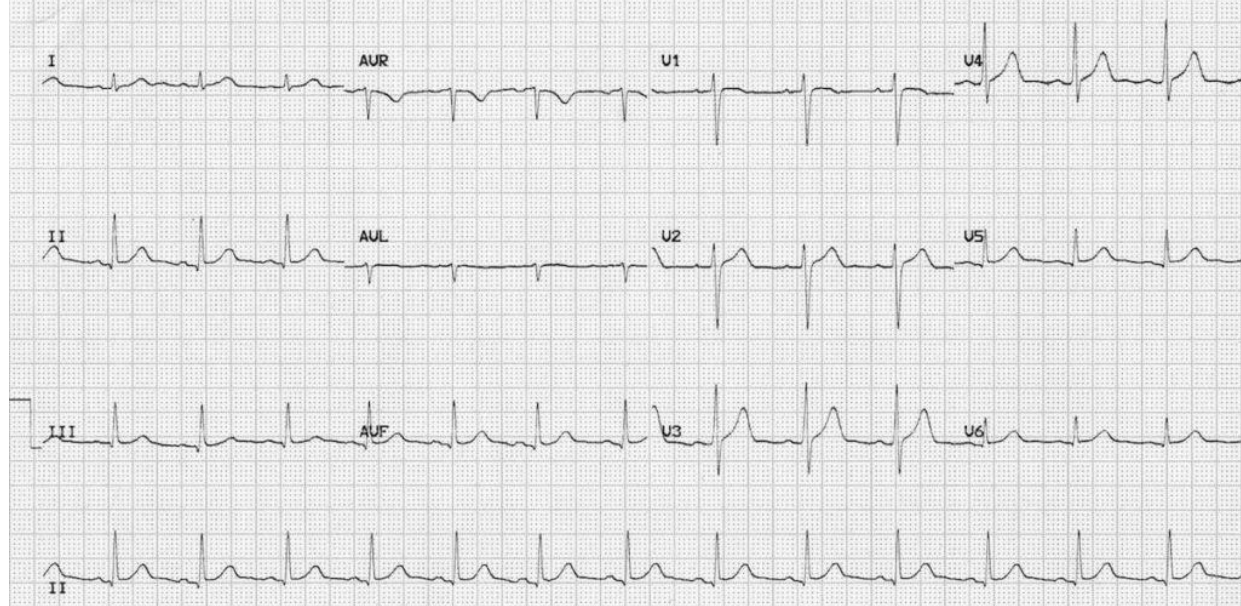
Appendix A: Laboratory Results

<u>CBC</u> WBC 12.5 Hgb 133 Plt 270 <u>Lytes</u> Na 136 K 3.6 Cl 97 HCO ₃ 24 Urea 8.1 Cr 143 Glucose 6.9 <u>Extended Lytes</u> Ca 1.33 Mg 0.70 PO ₄ 0.78 Albumin 38 <u>VBG</u> pH 7.38 pCO ₂ 36 HCO ₃ 25 Lactate 1.2	<u>Cardiac/Coags</u> Trop 4 INR 1.0 aPTT 30 <u>Biliary</u> AST 44 ALT 38 GGT 40 ALP 97 Bili 7 Lipase 24 <u>Tox</u> EtOH neg ASA neg Tylenol neg Osmols 304
---	---

Hypertensive Encephalopathy

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!



Normal Sinus Rhythm (From Life in the Fast Lane Blog)

<https://litfl.com/normal-sinus-rhythm-ecg-library/>

CT Head Report

Clinical details

Acute confusion

Technique

CT head without contrast

Findings

No mass, hemorrhage or hydrocephalus. Basal ganglia and posterior fossa structures are normal. No established major vessel vascular territory infarct. No intra or extra axial collection. The basal cisterns and foramen magnum are patent. The air cells of the petrous temporal bone are non-opacified. No fracture demonstrated.

Conclusion

Normal study

Hypertensive Encephalopathy

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

1. Recognize the unstable floor patient and call for help
2. Review differential diagnosis and work-up of confusion
3. Clear communication with team/maintain situational awareness as the leader
4. Discuss logistics of transferring to a critical care bed (ie RACE nurse is key)
5. Define hypertensive emergency: “a high blood pressure AND signs of end-organ dysfunction”
 - No formal numbers... but generally >180/110
6. Describe an approach to treating HTN emergencies
 - Many agents available... use what you are comfortable with, target a reduction in SBP of roughly 25% in first 1-2hours.
 - Arterial line and monitored setting

References

1. <https://emcrit.org/ibcc/hypertensive-emergency/>
- 2.
- 3.

